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GitHub Link	https://github.com/chaitramanohar/MLA0402-DEEP-LEARNING

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I T.CHAITRA(192425217)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T.chaitra

Name of the student:T.CHAITRA

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Question

Answer

Hospitals collect thousands of ECG (Electrocardiogram) signals every day. Since these records are **unlabeled**, supervised learning cannot be applied. A suitable solution is **Autoencoders**, which perform **unsupervised learning** by learning the normal patterns in ECG signals.

- They do not require labeled data.
- They learn hidden representations of normal ECG signals.
- When an abnormal ECG is given, reconstruction becomes poor.
- The reconstruction error is used to detect anomalies.

Workflow

1

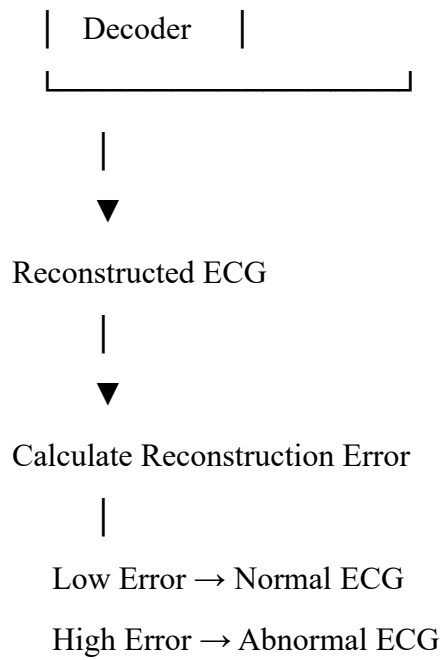


(Normalization & Noise Removal)



Encoder

| Compressed Code |



Steps

Step 1

Collect unlabeled ECG records.

Step 2

Normalize the ECG signals.

Step 3

Train the autoencoder using normal ECG patterns.

Step 4

The encoder compresses the ECG into important features.

Step 5

The decoder reconstructs the original ECG.

Step 6

Calculate reconstruction error.

- Small Error → Healthy ECG
- Large Error → Abnormal ECG

Advantages

- No labeled dataset required.

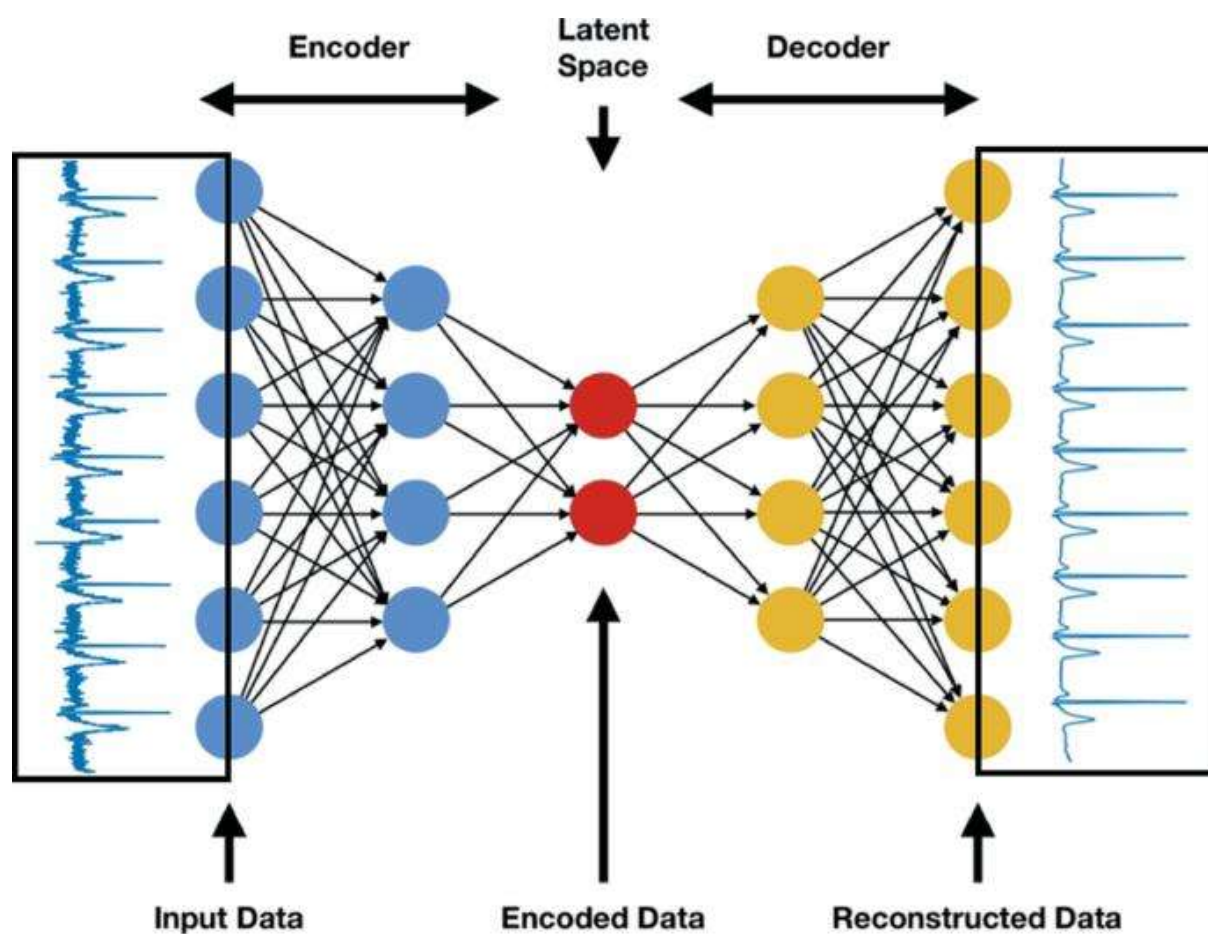
- Learns meaningful ECG features automatically.
- Detects rare diseases effectively.
- Reduces manual diagnosis effort.

Applications

- Heart disease detection
- Arrhythmia detection
- Patient monitoring
- Medical anomaly detection

Conclusion

Autoencoders are the best choice because they learn normal ECG patterns without labels and identify abnormal signals through reconstruction error, making them highly effective for medical anomaly detection.



2. Auto Encoders, Representation Learning

Question

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Introduction

Retail companies collect customer information such as:

- Age
- Income
- Products purchased
- Shopping frequency
- Payment methods
- Online activity

These datasets contain many features. Autoencoders reduce dimensionality while preserving useful information.

Why Autoencoders?

Autoencoders perform **representation learning**.

Instead of manually selecting features, they automatically learn a compact representation of customer behavior.

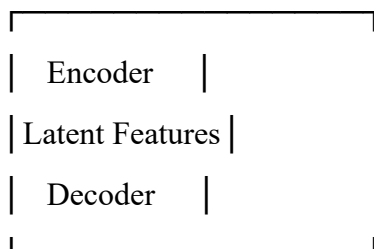
Workflow

Customer Purchase Dataset

|



Autoencoder



|



Compressed Customer Features

|



Customer Segmentation

(K-Means / Clustering)

Working

Step 1

Input customer purchase dataset.

Step 2

Encoder compresses hundreds of features into fewer meaningful features.

Example

100 Features



10 Features

These 10 features capture customer purchasing behavior.

Step 3

Decoder reconstructs the original data.

Step 4

Training minimizes reconstruction error.

Step 5

Use compressed features for clustering.

Customers are grouped into:

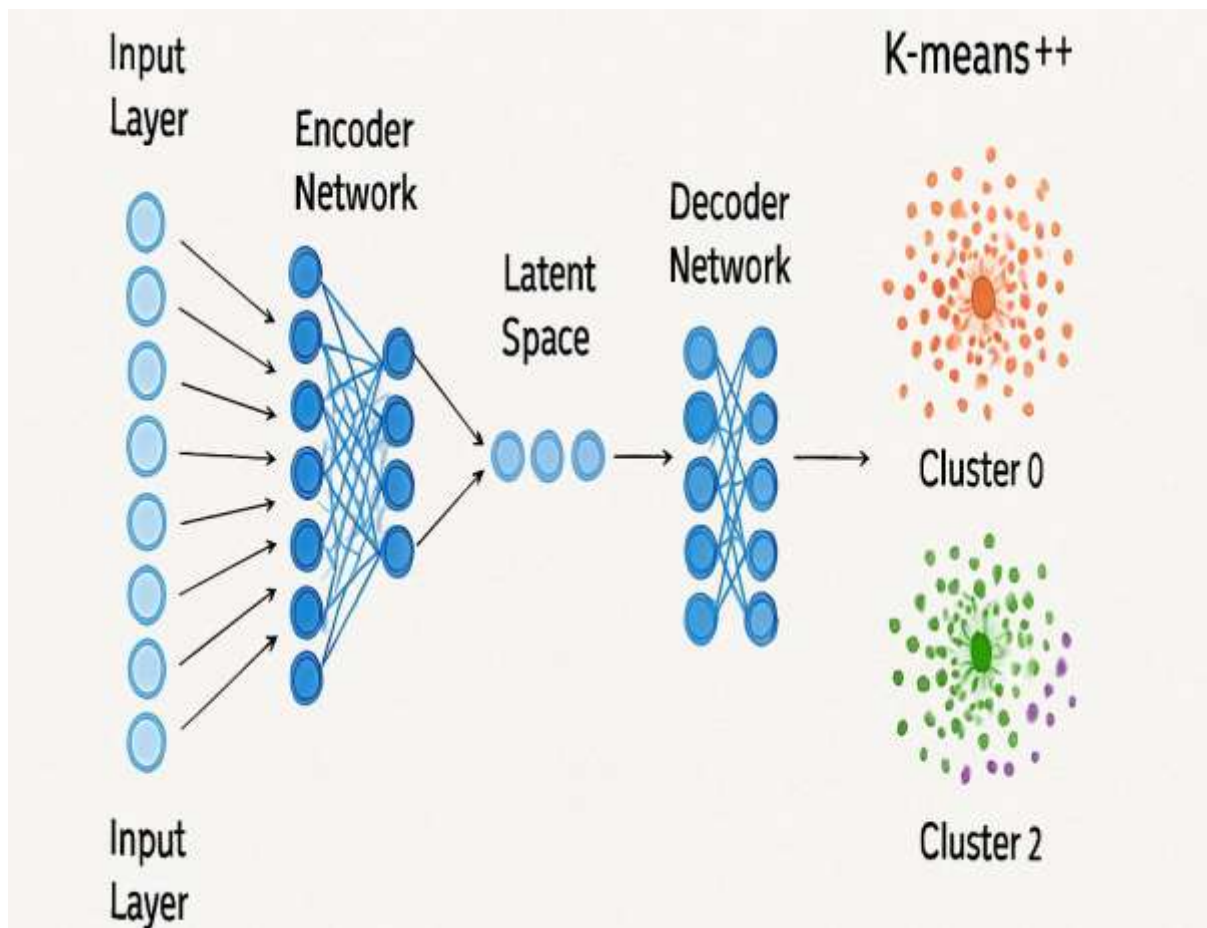
- Premium customers
- Regular buyers
- Seasonal buyers
- Discount shoppers

Benefits

- Reduces storage requirements.
- Removes noise.
- Faster clustering.
- Better customer segmentation.
- Learns useful hidden patterns.

Conclusion

Autoencoders efficiently reduce data dimensionality while preserving important customer information, making customer segmentation faster and more accurate.



3. Width and Depth of Neural Networks, Activation Functions

Question

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Introduction

A shallow neural network contains only one or two hidden layers.

Facial recognition is a complex task requiring extraction of low-level and high-level image features.

Recommended Changes

Increase Network Depth

Instead of:

Input

↓

Hidden Layer

↓

Output

Use

Input

↓

Conv Layer

↓

Conv Layer

↓

Conv Layer

↓

Fully Connected

↓

Output

Deep networks learn:

- Edges
- Eyes
- Nose
- Face structure
- Facial identity

Increase Width

Use more neurons per layer.

Example

Old

Input

↓

64 Neurons

New

Input

↓

256 Neurons

More neurons improve feature learning.

Use Better Activation Functions

Instead of Sigmoid,

Use **ReLU**.

$\text{ReLU}(x) = \max(0, x)$

Advantages:

- Faster learning.
- Avoids vanishing gradients.
- Efficient computation.

Hidden layers:

✓ ReLU

Output layer:

- Softmax (multiple faces)
- Sigmoid (binary face recognition)

Improved Architecture

Input Image

↓

Convolution Layer

↓

ReLU

↓

Pooling

↓

Convolution

↓

ReLU

↓

Fully Connected

↓

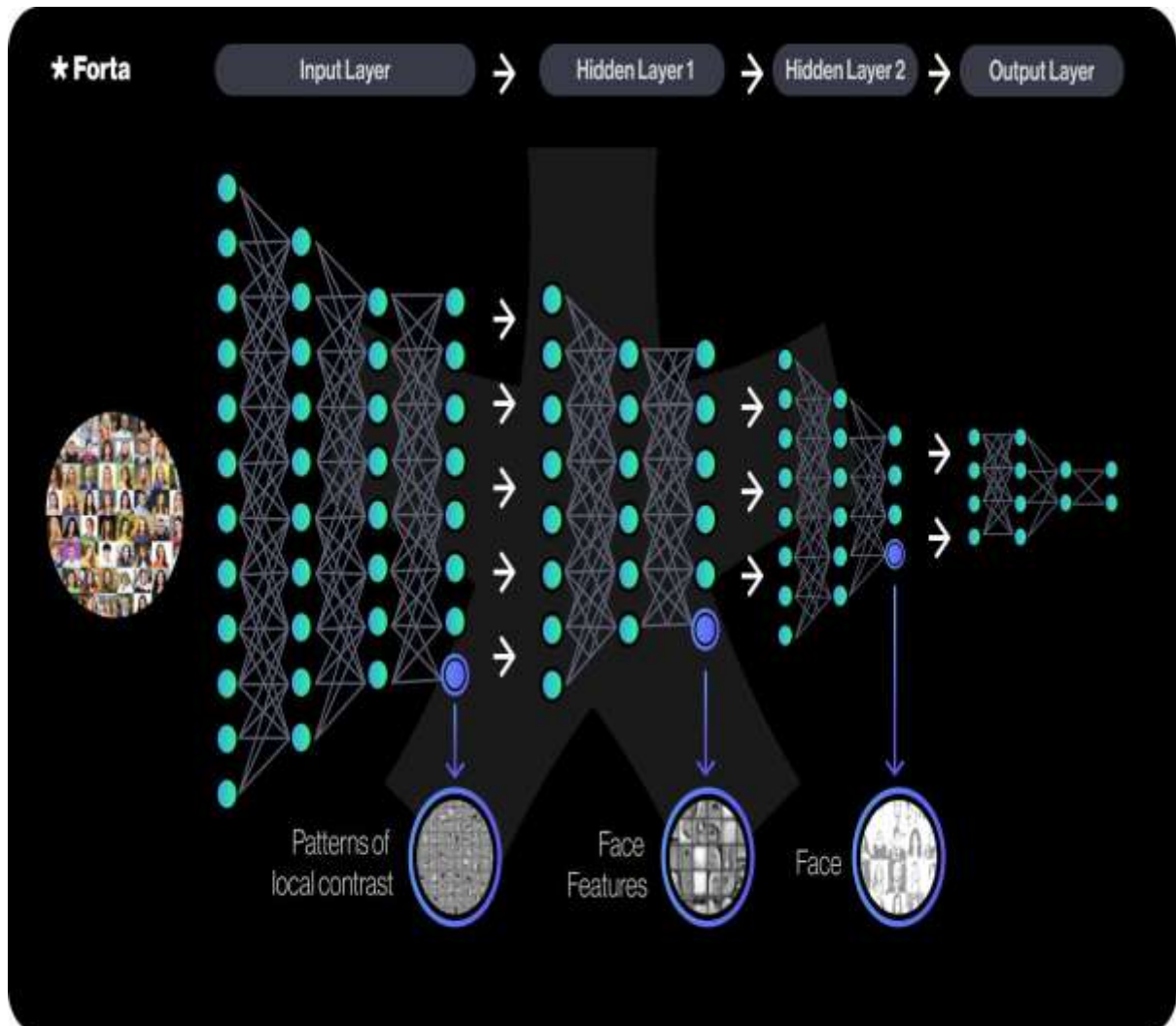
Softmax

Benefits

- Learns hierarchical facial features.
- Higher recognition accuracy.
- Faster convergence.
- Better generalization.

Conclusion

Increasing network depth and width while using ReLU activation enables the model to learn complex facial features effectively, significantly improving recognition performance.



4. Restricted Boltzmann Machines (RBMs), Unsupervised Training

Question

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Introduction

A movie streaming platform stores:

- Movies watched
- Ratings
- Watch time
- Viewing history

Since no labels are available, RBMs are suitable for unsupervised recommendation.

What is RBM?

Restricted Boltzmann Machine is a neural network with:

- Visible layer
- Hidden layer

There are no connections within the same layer.

Architecture

Hidden Layer

○ ○ ○ ○ ○

| | | |

Visible Layer

Movie1 Movie2 Movie3 Movie4 Movie5

Visible neurons represent movies.

Hidden neurons represent user preferences.

Workflow

Viewing History



RBM Training



Hidden Feature Learning



Predict Missing Ratings



Recommend Movies

Working

Step 1

Input user-movie interaction matrix.

Example

User **Movie A** **Movie B** **Movie C**

U1 Yes No Yes

Step 2

Train RBM.

Step 3

Hidden neurons learn preferences.

Example

Action Lover

Comedy Lover

Sci-Fi Lover

Step 4

Predict unseen movies.

If a user likes action movies,

Recommend

- Action Movie 1
- Action Movie 2
- Action Movie 3

Advantages

- No labeled data required.
- Learns hidden user interests.
- Handles sparse data.
- Improves personalized recommendations.

Applications

- Netflix
- Amazon
- Spotify
- YouTube

Conclusion

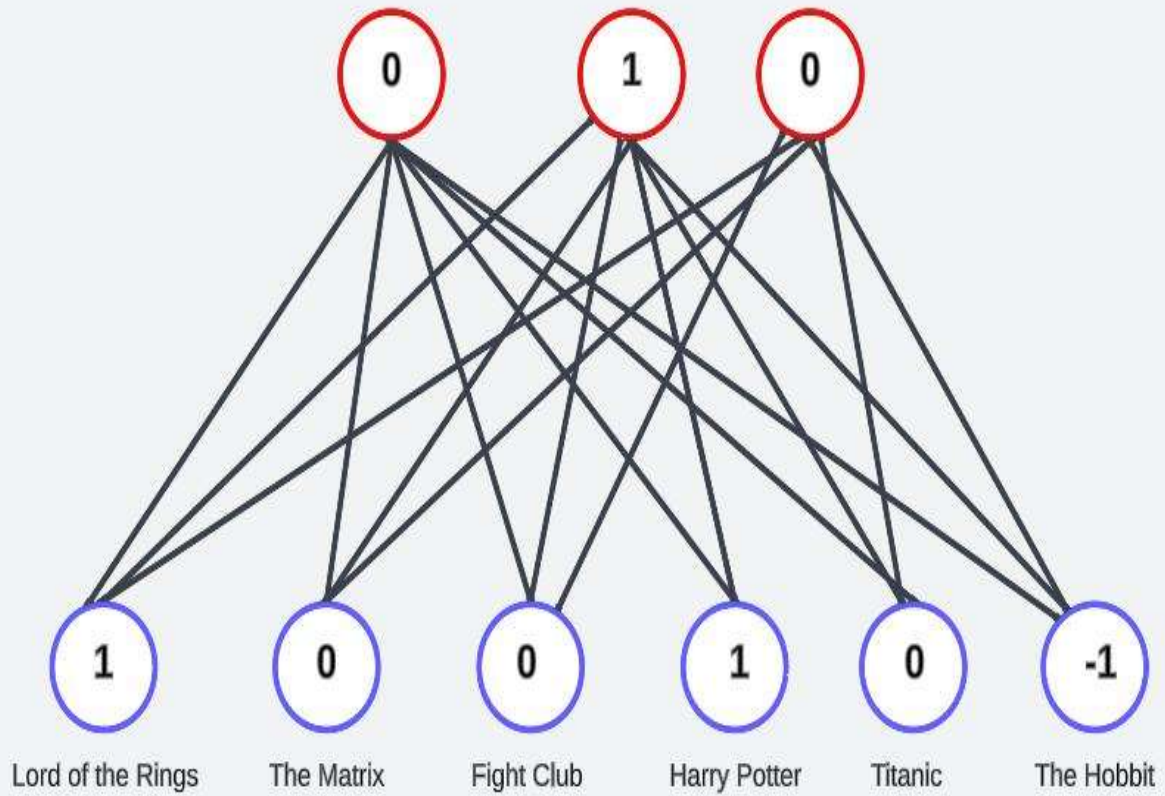
RBMs learn hidden user preferences from viewing history and predict unseen movies, making them an effective unsupervised recommendation technique.

Hidden Layer

$P = 0.354$

$P = 0.838$

$P = 0.838$



Visible Layer

5. Deep Learning Applications, Representation Learning

Question

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Introduction

Manufacturing machines continuously generate sensor readings such as:

- Temperature
- Pressure
- Vibration
- Speed
- Voltage

The goal is to automatically detect defective products in real time.

Proposed Solution

Use a **deep neural network with an Autoencoder** for representation learning and anomaly detection.

Workflow

Sensor Readings

|



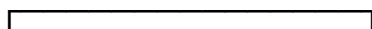
Data Preprocessing

(Cleaning & Normalization)

|

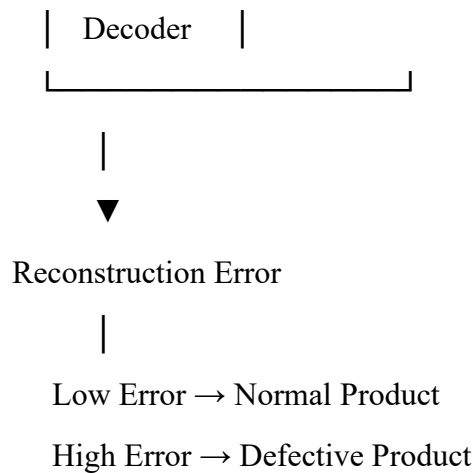


Autoencoder



| Encoder |

| Latent Features |



Working

Step 1

Collect sensor readings every second.

Step 2

Normalize and preprocess the data.

Step 3

Train the autoencoder using normal production data.

Step 4

The encoder learns compact representations of normal machine behavior.

Step 5

The decoder reconstructs the original sensor readings.

Step 6

Measure reconstruction error:

- **Low error:** Product is normal.
- **High error:** Product is likely defective.

Step 7

Generate an alert so defective products can be removed from the production line.

Benefits

- No need for labeled defect data.
- Learns normal operating patterns automatically.

- Detects faults in real time.
- Reduces production losses and maintenance costs.
- Improves product quality and reliability.

Applications

- Smart factories
- Predictive maintenance
- Industrial IoT
- Quality inspection

Conclusion

A deep learning solution based on representation learning with an autoencoder can effectively detect defective products by learning normal sensor patterns and identifying anomalies through reconstruction error, making it well suited for real-time manufacturing environments.

